



Faculté des Sciences et Technologies (FST)

Rapport du travail de Laboratoire N° 4_Réseaux II

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Niveau : L3

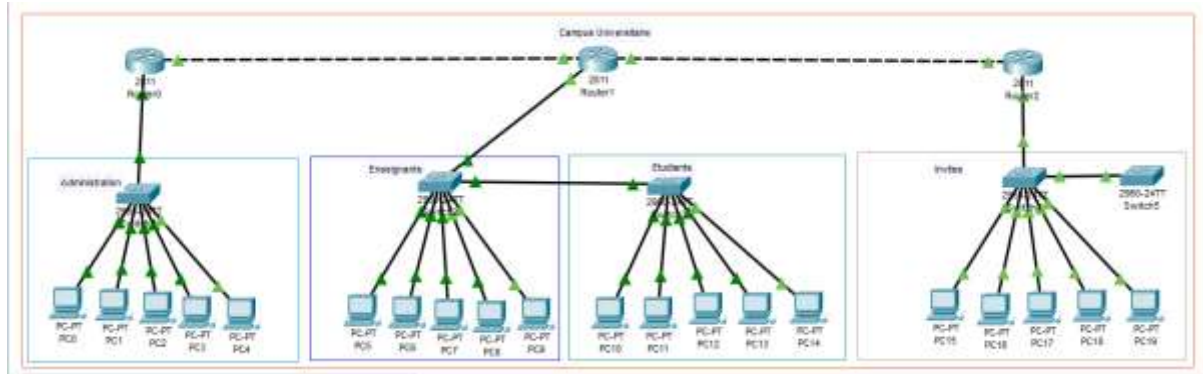
Le 20 avril 2026

Objectifs :

- Comprendre le rôle des protocoles de routage dynamique
- Configurer RIP v2, OSPF, EIGRP
- Comparer leur comportement
- Tester la convergence et la connectivité
- Simuler des pannes réseau

Création de la topologie réseau

- 3 Routeurs (R1, R2, R3)
- 5 Switches (S1 à S5)
- 25 PC



1-Configuration des interfaces sur les routeurs

Router0

Physical Config **CLI** Attributes

IOS Command Line Interface

```
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!

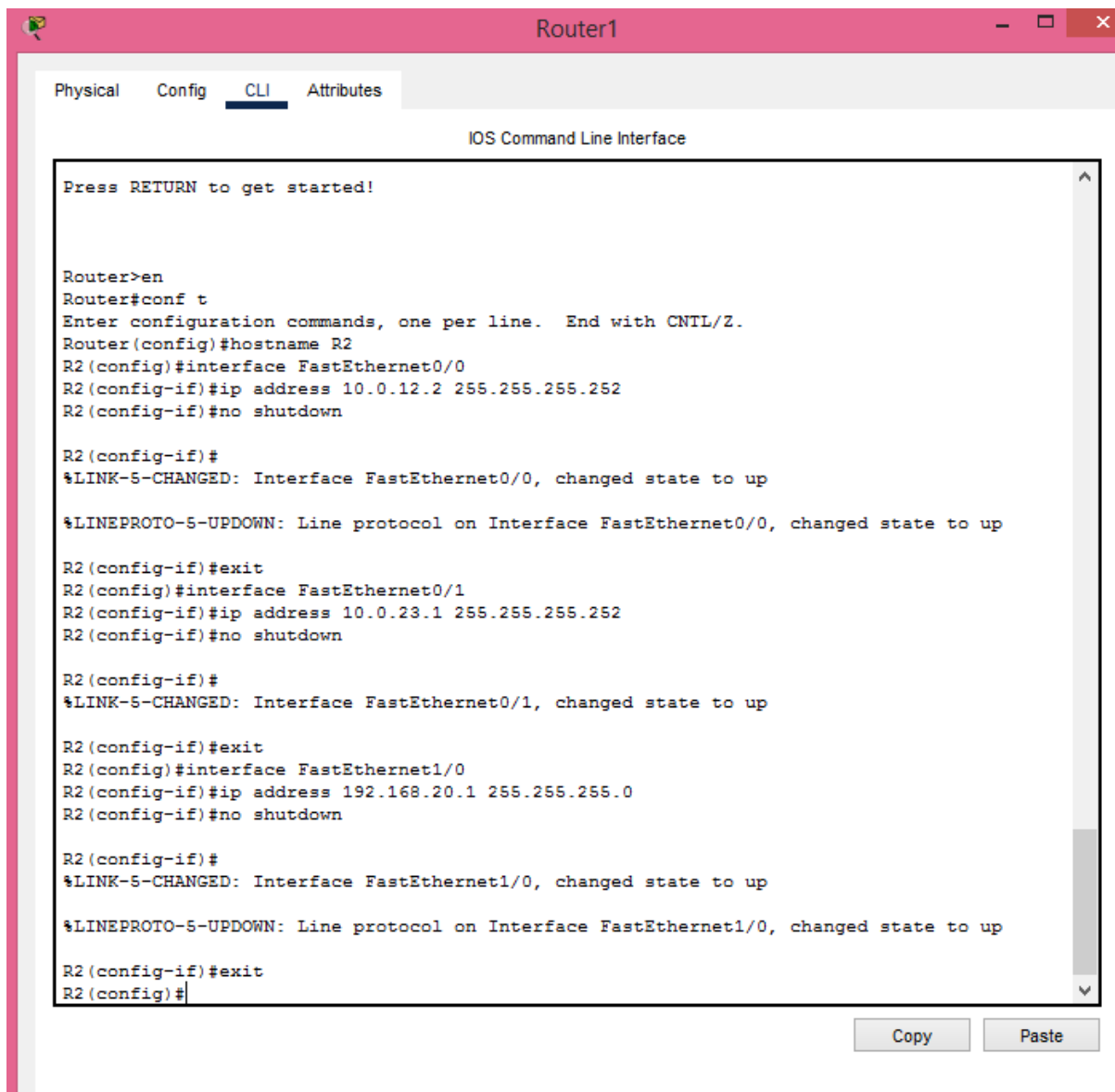
Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 10.0.12.1 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
interface FastEthernet0/1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
R1(config)#
```

Copy Paste



Router2

Physical Config **CLI** Attributes

IOS Command Line Interface

DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R3
R3(config)#interface FastEthernet0/0
R3(config-if)#ip address 10.0.23.2 255.255.255.252
R3(config-if)#no shutdown

R3(config-if)#
%LINK-S-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R3(config-if)#exit
R3(config)#interface FastEthernet0/1
R3(config-if)#ip address 192.168.30.1 255.255.255.0
R3(config-if)#no shutdown

R3(config-if)#
%LINK-S-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

R3(config-if)#exit
R3(config)#

2-Configuration des protocoles de routage dynamique

- Configuration de RIP v2

```
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#no auto-summary
R1(config-router)#network 192.168.10.0
R1(config-router)#network 10.0.12.0
R1(config-router)#exit
R1(config)#
```

```

R2>en
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#no auto-summary
R2(config-router)#network 192.168.20.0
R2(config-router)#network 10.0.12.0
R2(config-router)#network 10.0.23.0
R2(config-router)#exit
R2(config)#

```

```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#no auto-summary
R3(config-router)#network 192.168.30.0
R3(config-router)#network 10.0.23.0
R3(config-router)#exit
R3(config)#

```

- Configuration de OSPF

```

R1(config)#router ospf 1
R1(config-router)#network 192.168.10.0 0.0.0.255 area 0
R1(config-router)#network 10.0.12.0 0.0.0.3 area 0
R1(config-router)#exit
R1(config)#

```

```

R2(config)#router ospf 1
R2(config-router)#network 192.168.20.0 0.0.0.255 area 0
R2(config-router)#network 10.0.12.0 0.0.0.3 area 0
R2(config-router)#
00:43:36: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.1 on FastEthernet0/0 from LOADING to
FULL, Loading Done
network 10.0.23.0 0.0.0.3 area 0
R2(config-router)#exit
R2(config)#

```

```

R3(config)#router ospf 1
R3(config-router)#network 192.168.30.0 0.0.0.255 area 0
R3(config-router)#network 10.0.23.0 0.0.0.3 area 0
R3(config-router)#exit
R3(config)#

```

- Configuration de EIGRP

```

R1(config)#router eigrp 100
R1(config-router)#network 192.168.10.0
R1(config-router)#network 10.0.12.0
R1(config-router)#no auto-summary
R1(config-router)#exit
R1(config)#

```

```

R2(config)#router eigrp 100
R2(config-router)#network 192.168.20.0
R2(config-router)#network 10.0.12.0
R2(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.12.1 (FastEthernet0/0) is up: new
adjacency
network 10.0.23.0
R2(config-router)#no auto-summary
R2(config-router)#exit
R2(config)#

```

```

R3(config)#router eigrp 100
R3(config-router)#network 192.168.30.0
R3(config-router)#network 10.0.23.0
R3(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.23.1 (FastEthernet0/0) is up: new
adjacency
no auto-summary
R3(config-router)#exit
R3(config)#

```

❖ Configuration du serveur DHCP sur les routeurs

```

R1(config)#ip dhcp pool VLAN10
R1(dhcp-config)#network 192.168.10.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.10.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#exit
R1(config)#

```

```

R2(config)#ip dhcp pool VLAN20
R2(dhcp-config)#network 192.168.20.0 255.255.255.0
R2(dhcp-config)#default-router 192.168.20.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
R2(config)#ip dhcp pool VLAN30
R2(dhcp-config)#network 192.168.30.0 255.255.255.0
R2(dhcp-config)#default-router 192.168.30.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
R2(config)#

```

```

R3(config)#ip dhcp pool VLAN40
R3(dhcp-config)#network 192.168.40.0 255.255.255.0
R3(dhcp-config)#192.168.40.1
^
% Invalid input detected at '^' marker.

R3(dhcp-config)#default-router 192.168.40.1
R3(dhcp-config)#dns-server 8.8.8.8
R3(dhcp-config)#exit
R3(config)#

```


❖ Création des VLANs sur chaque switch

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#vlan 10
S1(config-vlan)#name Administration
S1(config-vlan)#exit
S1(config)#vlan 20
S1(config-vlan)#name Enseignants
S1(config-vlan)#exit
S1(config)#vlan 30
S1(config-vlan)#name Etudiants
S1(config-vlan)#exit
S1(config)#vlan 40
S1(config-vlan)#name Invites
S1(config-vlan)#exit
S1(config)#
```

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S2
S2(config)#vlan 10
S2(config-vlan)#name Administration
S2(config-vlan)#exit
S2(config)#vlan 20
S2(config-vlan)#name Enseignants
S2(config-vlan)#exit
S2(config)#vlan 30
S2(config-vlan)#name Etudiants
S2(config-vlan)#exit
S2(config)#vlan 40
S2(config-vlan)#name Invites
S2(config-vlan)#exit
S2(config)#
```

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S3
S3(config)#vlan 10
S3(config-vlan)#name Administration
S3(config-vlan)#exit
S3(config)#vlan 20
S3(config-vlan)#name Enseignants
S3(config-vlan)#exit
S3(config)#vlan 30
S3(config-vlan)#name Etudiants
S3(config-vlan)#exit
S3(config)#vlan 40
S3(config-vlan)#name Invites
S3(config-vlan)#exit
S3(config)#
```

```

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S4
S4(config)#vlan 10
S4(config-vlan)#name Administration
S4(config-vlan)#exit
S4(config)#vlan 20
S4(config-vlan)#name Enseignants
S4(config-vlan)#exit
S4(config)#vlan 30
S4(config-vlan)#name Etudiants
S4(config-vlan)#exit
S4(config)#vlan 40
S4(config-vlan)#name Invites
S4(config-vlan)#exit
S4(config)#

```

```

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S5
S5(config)#vlan 10
S5(config-vlan)#name Administration
S5(config-vlan)#exit
S5(config)#vlan 20
S5(config-vlan)#name Enseignants
S5(config-vlan)#exit
S5(config)#vlan 30
S5(config-vlan)#name Etudiants
S5(config-vlan)#exit
S5(config)#vlan 40
S5(config-vlan)#name Invites
S5(config-vlan)#exit
S5(config)#

```

❖ Assignment des ports aux VLANs

```

S1>en
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#vlan 10
S1(config-vlan)#name Administration
S1(config-vlan)#interface range FastEthernet0/1-5
S1(config-if-range)#switchport mode access
S1(config-if-range)#switchport access vlan 10
S1(config-if-range)#no shutdown
S1(config-if-range)#exit
S1(config)#

```

```

S1(config)#interface FastEthernet0/24
S1(config-if)#switchport mode trunk
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#

```

```
S2>en
S2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#vlan 20
S2(config-vlan)#Enseignants
^
% Invalid input detected at '^' marker.

S2(config-vlan)#name Enseignants
S2(config-vlan)#interface range FastEthernet0/1-5
S2(config-if-range)#switchport mode access
S2(config-if-range)#switchport access vlan 20
S2(config-if-range)#no shutdown
S2(config-if-range)#exit
S2(config)#
```

```
S2(config)#interface FastEthernet0/24
S2(config-if)#switchport mode trunk
S2(config-if)#no shutdown
S2(config-if)#exit
S2(config)#
```

```
S3>en
S3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S3(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with
S2 FastEthernet0/2 (20).

S3(config)#vlan 30
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with
S2 FastEthernet0/2 (20).

S3(config-vlan)#name Etudiants
S3(config-vlan)#interface range FastEthernet0/1-5
S3(config-if-range)#switchport mode access
S3(config-if-range)#switchport access vlan 30
S3(config-if-range)#no shutdown
S3(config-if-range)#exit
S3(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (30),
with S2 FastEthernet0/2 (20).
```

```
S3(config)#interface FastEthernet0/24
S3(config-if)#switchport mode trunk
S3(config-if)#no shutdown
S3(config-if)#exit
S3(config)#
```

```
S4>en
S4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S4(config)#vlan 40
S4(config-vlan)#name Invites
S4(config-vlan)#interface range FastEthernet0/1-5
S4(config-if-range)#switchport mode access
S4(config-if-range)#switchport access vlan 40
S4(config-if-range)#no shutdown
S4(config-if-range)#exit
S4(config)#
```

```
S4(config)#interface
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (40),
with S5 FastEthernet0/1 (1).
FastEthernet0/24
S4(config-if)#switchport mode trunk
S4(config-if)#no shutdown
S4(config-if)#exit
S4(config)#
```

❖ Configuration du routage inter-VLAN sur les routeurs

```
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface FastEthernet0/1.10
R1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.10, changed state to up
encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.0
% 192.168.10.0 overlaps with FastEthernet0/1
R1(config-subif)#no shutdown
R1(config-subif)#exit
R1(config)#
```

```
R2>en
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#interface FastEthernet0/1.20
R2(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.20, changed state to up
encapsulation dot1Q 20
R2(config-subif)#ip address 192.168.20.1 255.255.255.0
% 192.168.20.0 overlaps with FastEthernet1/0
R2(config-subif)#no shutdown
R2(config-subif)#interface FastEthernet0/1.30
R2(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.30, changed state to up
encapsulation dot1Q 30
R2(config-subif)#ip address 192.168.30.1 255.255.255.0
R2(config-subif)#no shutdown
R2(config-subif)#exit
R2(config)#
```

```
R3>en
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface FastEthernet0/1.40
R3(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.40, changed state to up
encapsulation dot1Q 40
R3(config-subif)#
R3(config-subif)#ip address 192.168.40.1 255.255.255.0
R3(config-subif)#no shutdown
R3(config-subif)#exit
R3(config)#
```

❖ Test

✓ Vérification les VLANs :

```
S1>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Administration	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5
20	Enseignants	active	
30	Etudiants	active	
40	Invites	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
S1>
```

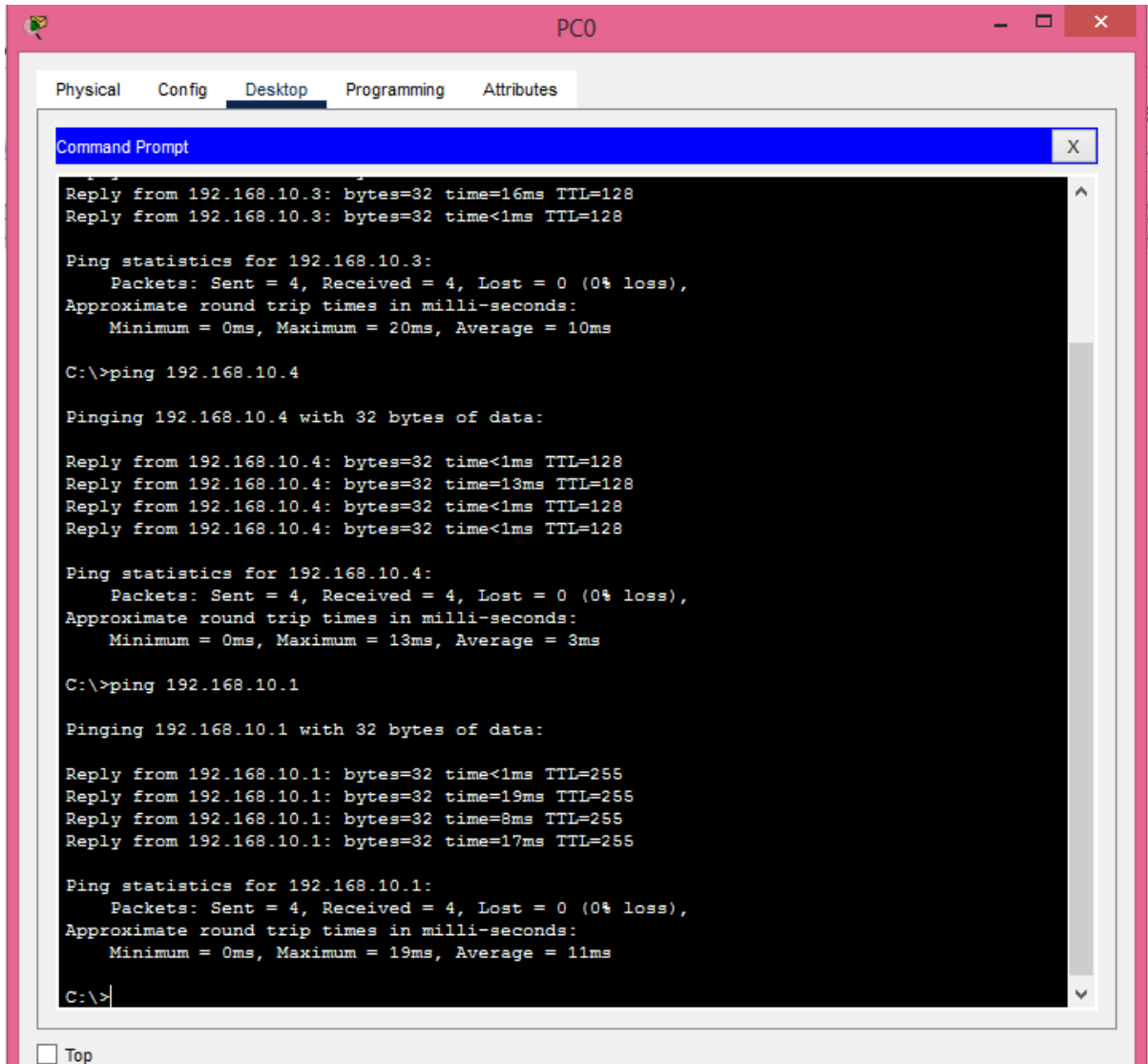
```
S4>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Administration	active	
20	Enseignants	active	
30	Etudiants	active	
40	Invites	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
S4>
```

✓ ping intra-VLAN

Sur le vlan 10 (PC0 – R1, PC1 et PC2)



The screenshot shows a PC0 desktop environment with a Command Prompt window open. The window displays the results of three ping commands executed from the PC. The first command is for 192.168.10.3, showing a successful ping with 4 packets sent and received, 0% loss, and an average round trip time of 10ms. The second command is for 192.168.10.4, also showing a successful ping with 4 packets sent and received, 0% loss, and an average round trip time of 3ms. The third command is for 192.168.10.1, showing a successful ping with 4 packets sent and received, 0% loss, and an average round trip time of 11ms. The Command Prompt window has a title bar that says 'PC0' and a menu bar with 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is selected. The Command Prompt window has a title bar that says 'Command Prompt' and a close button 'X'.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Reply from 192.168.10.3: bytes=32 time=16ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 10ms

C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=13ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

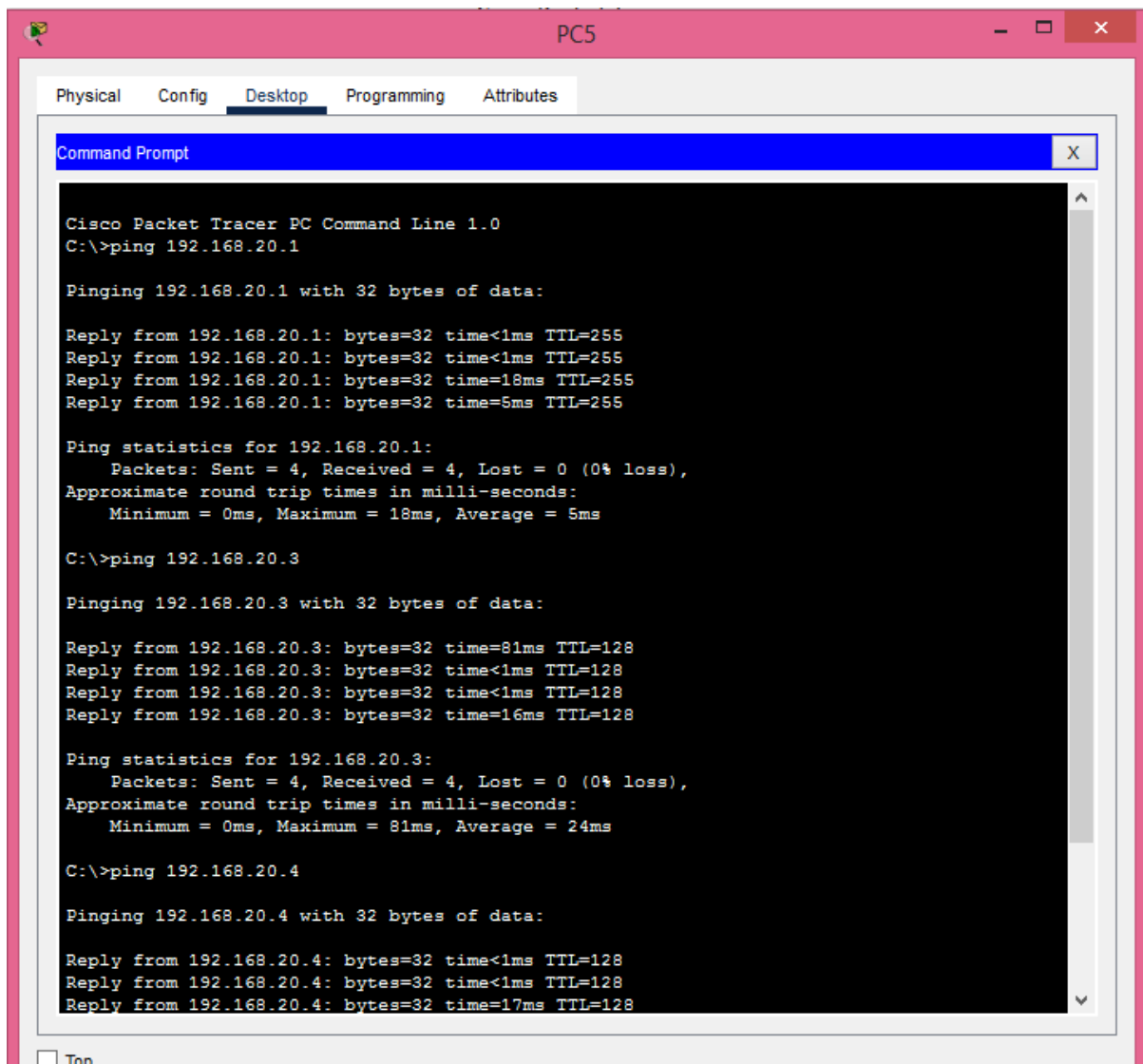
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=19ms TTL=255
Reply from 192.168.10.1: bytes=32 time=8ms TTL=255
Reply from 192.168.10.1: bytes=32 time=17ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 19ms, Average = 11ms

C:\>
```

☐ Top

Sur le vlan 20 (PC5 – R2, PC6 et PC7)



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=18ms TTL=255
Reply from 192.168.20.1: bytes=32 time=5ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 5ms

C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time=81ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time=16ms TTL=128

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 81ms, Average = 24ms

C:\>ping 192.168.20.4

Pinging 192.168.20.4 with 32 bytes of data:

Reply from 192.168.20.4: bytes=32 time<1ms TTL=128
Reply from 192.168.20.4: bytes=32 time<1ms TTL=128
Reply from 192.168.20.4: bytes=32 time=17ms TTL=128
```

✓ Ping inter-VLAN

The screenshot shows a PC0 Desktop window with a Command Prompt open. The Command Prompt displays the results of ping commands for two destinations: 192.168.20.2 and 192.168.20.5. The results show that all four packets sent to each destination were received, with 0% loss. The approximate round trip times in milliseconds are provided for each destination.

```

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=15ms TTL=126
Reply from 192.168.20.2: bytes=32 time=18ms TTL=126
Reply from 192.168.20.2: bytes=32 time=1ms TTL=126
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 8ms

C:\>ping 192.168.20.5

Pinging 192.168.20.5 with 32 bytes of data:

Reply from 192.168.20.5: bytes=32 time=12ms TTL=126
Reply from 192.168.20.5: bytes=32 time=49ms TTL=126
Reply from 192.168.20.5: bytes=32 time<1ms TTL=126
Reply from 192.168.20.5: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 49ms, Average = 15ms

C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:
  
```

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC0	PC5	ICMP		0.000	N	0	(edit)
	Successful	PC0	PC10	ICMP		0.000	N	1	(edit)
	Successful	PC5	PC10	ICMP		0.000	N	2	(edit)

✓ Routers

```
R1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C    10.0.12.0/30 is directly connected, FastEthernet0/0
L    10.0.12.1/32 is directly connected, FastEthernet0/0
D    10.0.23.0/30 [90/30720] via 10.0.12.2, 00:17:10, FastEthernet0/0
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, FastEthernet0/1
L    192.168.10.1/32 is directly connected, FastEthernet0/1
D    192.168.20.0/24 [90/30720] via 10.0.12.2, 00:17:10, FastEthernet0/0
D    192.168.30.0/24 [90/33280] via 10.0.12.2, 00:17:07, FastEthernet0/0
```

R1>

```
R2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.12.0/30 is directly connected, FastEthernet0/0
L    10.0.12.2/32 is directly connected, FastEthernet0/0
C    10.0.23.0/30 is directly connected, FastEthernet0/1
L    10.0.23.1/32 is directly connected, FastEthernet0/1
D    192.168.10.0/24 [90/30720] via 10.0.12.1, 00:51:31, FastEthernet0/0
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, FastEthernet1/0
L    192.168.20.1/32 is directly connected, FastEthernet1/0
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.30.0/24 is directly connected, FastEthernet0/1.30
L    192.168.30.1/32 is directly connected, FastEthernet0/1.30
```

--More--

```
R3>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```

10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
D    10.0.12.0/30 [90/30720] via 10.0.23.1, 00:52:02, FastEthernet0/0
C    10.0.23.0/30 is directly connected, FastEthernet0/0
L    10.0.23.2/32 is directly connected, FastEthernet0/0
D    192.168.10.0/24 [90/33280] via 10.0.23.1, 00:51:58, FastEthernet0/0
D    192.168.20.0/24 [90/30720] via 10.0.23.1, 00:52:02, FastEthernet0/0
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.30.0/24 is directly connected, FastEthernet0/1
L    192.168.30.1/32 is directly connected, FastEthernet0/1
192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.40.0/24 is directly connected, FastEthernet0/1.40
L    192.168.40.1/32 is directly connected, FastEthernet0/1.40
--More--

```

```
R1>show ip eigrp neighbors
IP-EIGRP neighbors for process 100
```

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO	Q Cnt	Seq Num
0	10.0.12.2	Fa0/0	11	00:19:29	40	1000	0	15

```
R1>show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.20.1	1	FULL/BDR	00:00:38	10.0.12.2	FastEthernet0/0

```
R1>show ip rip neighbor
```

❖ Questions TD

1. Pourquoi OSPF est plus efficace que RIP ?

R - OSPF est plus efficace parce qu'il utilise un algorithme de type "link-state" qui permet une convergence plus rapide et une meilleure scalabilité que RIP (distance-vector).

2. Quelle différence entre protocole classful et classless ?

R - Les protocoles classful (comme RIPv1) ne supportent pas le VLSM, tandis que les protocoles classless (comme RIPv2, OSPF, EIGRP) le supportent.

3. Pourquoi EIGRP est plus rapide ?

R - EIGRP est plus rapide parce qu'il utilise l'algorithme DUAL pour une convergence rapide et une gestion efficace des routes de secours.

4. Quel protocole choisir pour une grande entreprise et pourquoi ?

R - OSPF est souvent recommandé pour les grandes entreprises en raison de sa scalabilité, de sa rapidité de convergence et de sa capacité à gérer des réseaux complexes.

Conclusion

Ce travail a permis de configurer un réseau multi-VLAN avec routage dynamique (RIP, OSPF, EIGRP) et DHCP, en respectant une topologie de campus universitaire. Les tests de connectivité et de résilience ont validé la communication inter-VLAN et la convergence des protocoles.